E-Commerce Sales Data Analysis using Python

Project Documentation

E-Commerce Sales Data Analysis using Python

Cover Page

Field	Detail
	E-Commerce Sales Data Analysis using

Project Title	Python
Project Type	Industry Training Project — Data Analytics (Project 2)
Domain	E-Commerce / Retail Analytics
Author	Princy Khasa
Date	June 18, 2026
Version	1.0
Technology Stack	Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn

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1. Executive Summary

This project applies Exploratory Data Analysis (EDA) techniques in Python to a 1,200-record e-commerce order dataset to uncover patterns in product performance, customer acquisition channels, payment behavior, coupon effectiveness, and revenue trends. The analysis was performed entirely in a Jupyter Notebook using Pandas, NumPy, Matplotlib, and Seaborn, and produced 19 distinct visualizations supported by descriptive statistics and correlation analysis.

The most significant finding is that `UnitPrice` is the strongest numeric driver of order value (correlation of 0.717 with `TotalPrice`), ahead of `Quantity` (0.615) and `ItemsInCart` (0.393). At a category level, Chair, Printer, and Laptop are the top revenue-generating products, while Phone is the weakest performer. Instagram is the leading customer acquisition channel, and FREESHIP is the best-performing coupon by total revenue. Year-over-year revenue declined from ₹552,643.24 in 2023 to ₹480,235.87 in 2024 to ₹231,882.85 in 2025 (a partial year through June).

This document provides complete, file-verified documentation of every cleaning step, chart, statistic, and business insight produced during the project.

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2. Project Overview

The project analyzes order-level e-commerce transaction data to answer a core business question: what drives revenue, and where are the biggest opportunities for growth? Using Python’s data analysis stack, the project moves through data loading, light cleaning, descriptive statistics, univariate and bivariate visualization, correlation analysis, and time-series trend analysis, concluding with a set of actionable business recommendations.

The dataset itself was already largely clean prior to this stage of analysis — the EDA phase confirmed data quality (zero duplicate rows, only one column with missing values) before proceeding to insight generation.

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3. Business Problem Statement

An e-commerce business has accumulated 1,200 order records spanning January 2023 to June 2025 but has not yet translated this raw transactional data into structured insight. Without analysis, the business cannot answer fundamental questions such as which products to prioritize, which marketing channels deserve more budget, whether coupon discounts are working, or why revenue appears to be trending downward year over year. This project addresses that gap using Python-based exploratory data analysis.

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4. Project Objectives

- Load and inspect the e-commerce dataset for structure, size, and quality
- Identify and treat missing values and duplicate records
- Analyze the distribution of products, order statuses, and payment methods
- Detect outliers in pricing, quantity, and cart-size fields using box plots
- Examine relationships between order variables using scatter plots and a correlation heatmap
- Analyze revenue and order volume trends on a monthly basis across the full date range
- Evaluate product-level revenue, referral-channel performance, and coupon-level performance
- Translate every analytical finding into a specific, actionable business recommendation

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5. Scope of the Project

This project is limited to exploratory data analysis on a single, pre-existing dataset. It does not include predictive modeling, machine learning, database integration, or dashboard deployment. The analysis is confined to the 14 columns and 1,200 records present in the source file, covering the time period January 2023 through June 2025. No external data sources were merged or referenced.

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6. Dataset Description

Source: `Dataset_for_Data_Analytics.xlsx` (provided dataset, single sheet)

Number of Records: 1,200

Number of Columns: 14

Time Period Covered: January 1, 2023 to June 30, 2025

File Format: Excel (.xlsx)

Features and Data Types

Column	Data Type	Description
`OrderID`	object	Unique identifier for each order
`Date`	datetime64 (after conversion)	Date the order was placed

`CustomerID`	object	Unique identifier for the customer
`Product`	object	Product purchased (Monitor, Phone, Tablet, Chair, Printer, Laptop, Desk)
`Quantity`	int64	Units purchased in the order
`UnitPrice`	float64	Price per unit
`ShippingAddress`	object	Delivery address
`PaymentMethod`	object	Mode of payment (Debit Card, Credit Card, Online, Cash, Gift Card)
`OrderStatus`	object	Order status (Shipped, Delivered, Returned, Cancelled, Pending)
`TrackingNumber`	object	Shipment tracking identifier
`ItemsInCart`	int64	Number of items in the customer's cart at checkout
`CouponCode`	object	Coupon applied (Save10, Freeship, Winter15, or blank/No Coupon)
`ReferralSource`	object	Customer acquisition channel (Instagram, Google, Email, Facebook, Referral)
`TotalPrice`	float64	Final order value (Quantity × UnitPrice)

Description of Key Columns

`OrderID` serves as the unique transactional identifier and was verified to have zero duplicates across all 1,200 records. `CustomerID` identifies the purchasing customer; with 1,200 orders mapped to a smaller pool of unique customers, this field is central to understanding repeat-purchase behavior. `TotalPrice` is the primary revenue metric throughout the analysis, and was confirmed to be internally consistent with `Quantity × UnitPrice` for every record. `CouponCode` was the only column found to contain missing values (309 blank entries), which were treated as orders placed without any coupon during analysis.

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7. Technology Stack

Technology	Purpose / Reason for Use
Python	Core programming language used for the entire analysis; chosen for its mature data

	science ecosystem
Jupyter Notebook	Interactive environment allowing code, output, and visualizations to be developed and documented together
Pandas	Used for loading the dataset, data cleaning (null and duplicate handling), and groupby-based aggregation
NumPy	Used to support numerical operations underlying Pandas computations
Matplotlib	Used to render box plots, histograms, scatter plots, and trend line charts
Seaborn	Used for statistically-oriented visualizations including the correlation heatmap and distribution plots

No SQL, database engine, or BI tool was used in this notebook-based project; all analysis was performed directly on the in-memory Pandas DataFrame.

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8. Project Architecture / Workflow

```
Raw Dataset (Dataset_for_Data_Analytics.xlsx)
        ↓
Data Loading (pandas.read_excel)
        ↓
Data Cleaning (missing value treatment, duplicate check, date type
conversion)
        ↓
Exploratory Data Analysis
        ↓
Visualizations (19 charts: distributions, box plots, histograms, scatter
plots, heatmap, trends)
        ↓
Statistical Analysis (describe(), correlation matrix)
        ↓
Business Insights
        ↓
Recommendations
```

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9. Data Understanding

The dataset represents a single, flat table of e-commerce order transactions, with no relational joins or external lookup tables involved. Each row corresponds to exactly one order, identified uniquely by `OrderID`. The columns span four broad categories: identifiers (`OrderID`, `CustomerID`, `TrackingNumber`), categorical attributes (`Product`, `PaymentMethod`, `OrderStatus`, `ReferralSource`, `CouponCode`), numeric measures

(`Quantity`, `UnitPrice`, `ItemsInCart`, `TotalPrice`), and a single date field (`Date`). This structure made the dataset directly suitable for groupby-based aggregation and correlation analysis without requiring any schema redesign.

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10. Data Cleaning Process

The following cleaning steps were performed and verified directly within the notebook:

Step	Action Performed	Result
Missing value identification	Checked all columns for missing values	309 missing values found, all confined to `CouponCode`
Missing value treatment	Filled missing `CouponCode` values with `"No Coupon"`	0 missing values remaining in the dataset
Duplicate check	Counted fully duplicate rows using `df.duplicated().sum()`	0 duplicate rows found
Duplicate removal	Applied `df.drop_duplicates()` as a precautionary measure	No rows were removed, confirming the dataset was already free of duplicates
Data type conversion	Converted the `Date` column to `datetime` format using `pd.to_datetime()`	Enabled month- and year-based grouping for trend analysis
Data type review	Reviewed all column types using `df.dtypes` and `df.info()`	Confirmed all columns held appropriate types for analysis

No outlier removal, text standardization, or feature engineering steps were documented in the notebook beyond the missing-value and duplicate-handling steps described above. Outlier *detection* (without removal) was performed visually using box plots, covered in the EDA section below.

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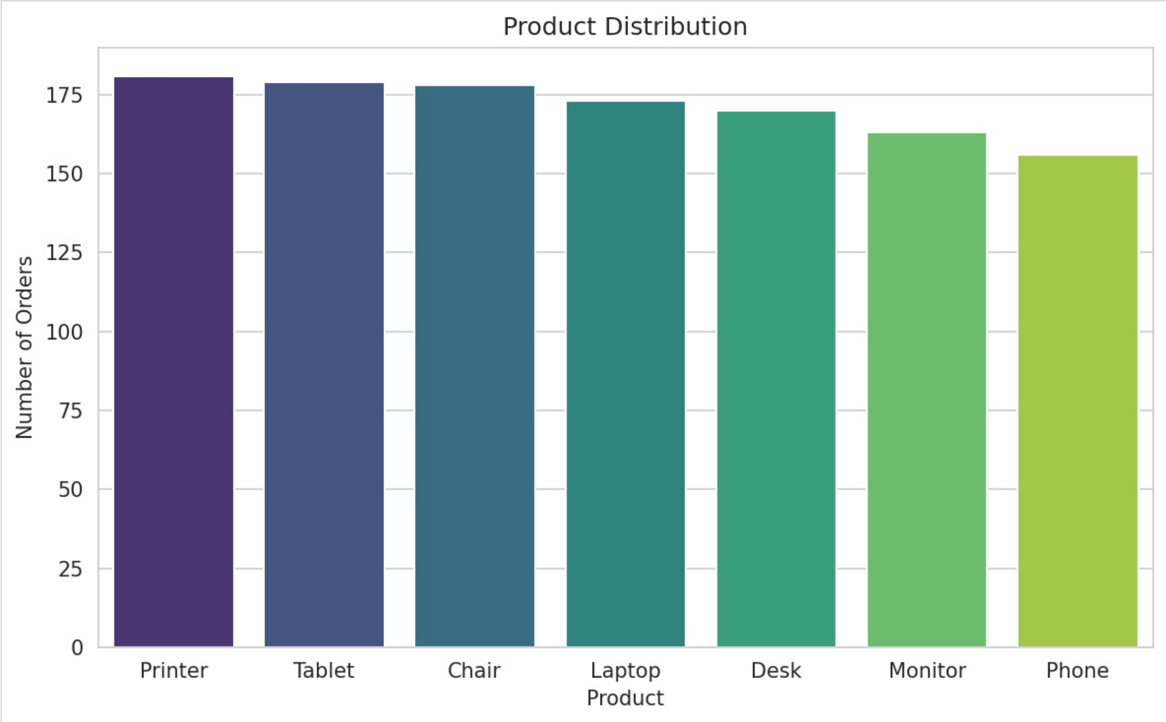
11. Exploratory Data Analysis (EDA)

Nineteen visualizations were produced in the notebook. Each is documented individually below with its purpose, observation, business insight, and recommendation. Screenshot placeholders are provided for each chart; these should be replaced with the actual exported images from the notebook.

Chart 1 — Product Distribution

Figure 1 – Product Distribution



Purpose: To visualize how many orders were placed for each product category. **Observation:** Printer received the highest number of orders among all products. **Business Insight:** Printer reflects strong and consistent customer demand relative to other categories. **Recommendation:** Maintain consistent stock availability for Printer and consider featuring it in promotional bundles.

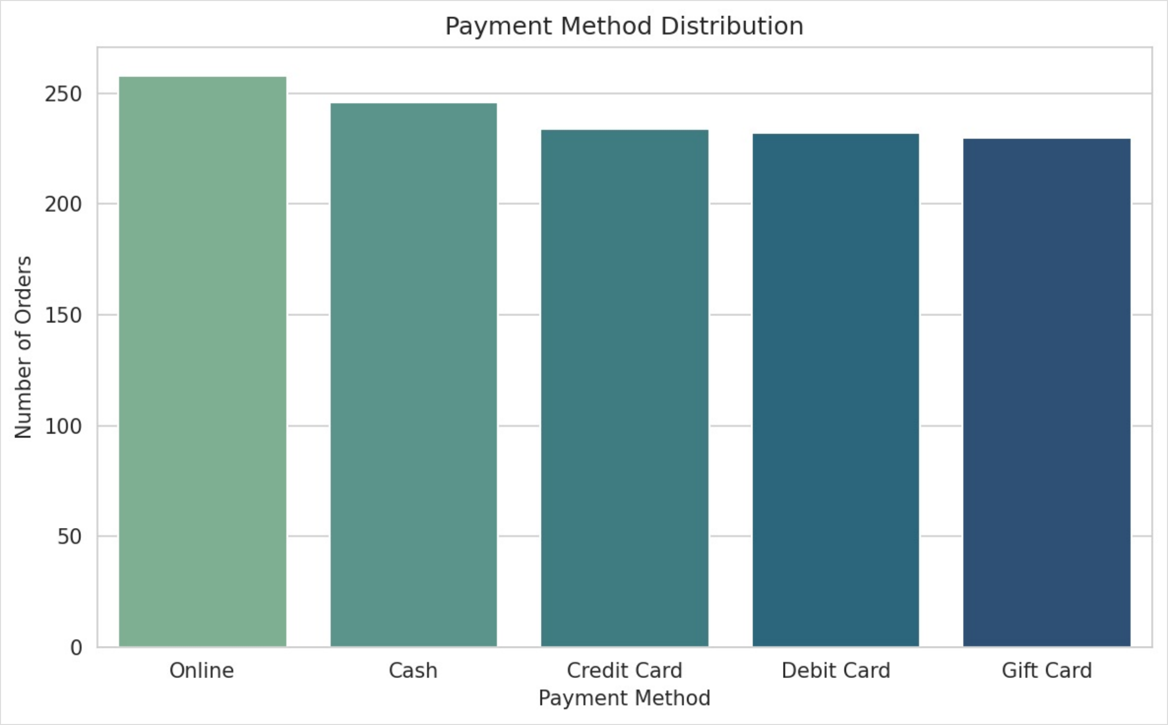
Chart 2 — Order Status Distribution

Figure 2 – Order Status Distribution

Purpose: To understand how orders are distributed across status categories. **Observation:** Cancelled orders (250) and Returned orders (247) together represent a large share of total orders, compared to 231 Delivered orders. **Business Insight:** The combined cancellation and return volume relative to successful deliveries points to potential issues in fulfillment or customer satisfaction. **Recommendation:** Investigate root causes such as delivery delays or product mismatches, and implement targeted process improvements.

Chart 3 — Payment Method Distribution

Figure 3 – Payment Method Distribution



Purpose: To visualize the distribution of orders across payment methods. **Observation:** Orders are spread across Debit Card, Credit Card, Online, Cash, and Gift Card with no single method dominating. **Business Insight:** Customer payment preference is diversified rather than concentrated in one channel. **Recommendation:** Continue supporting all current payment methods and monitor for future shifts in preference.

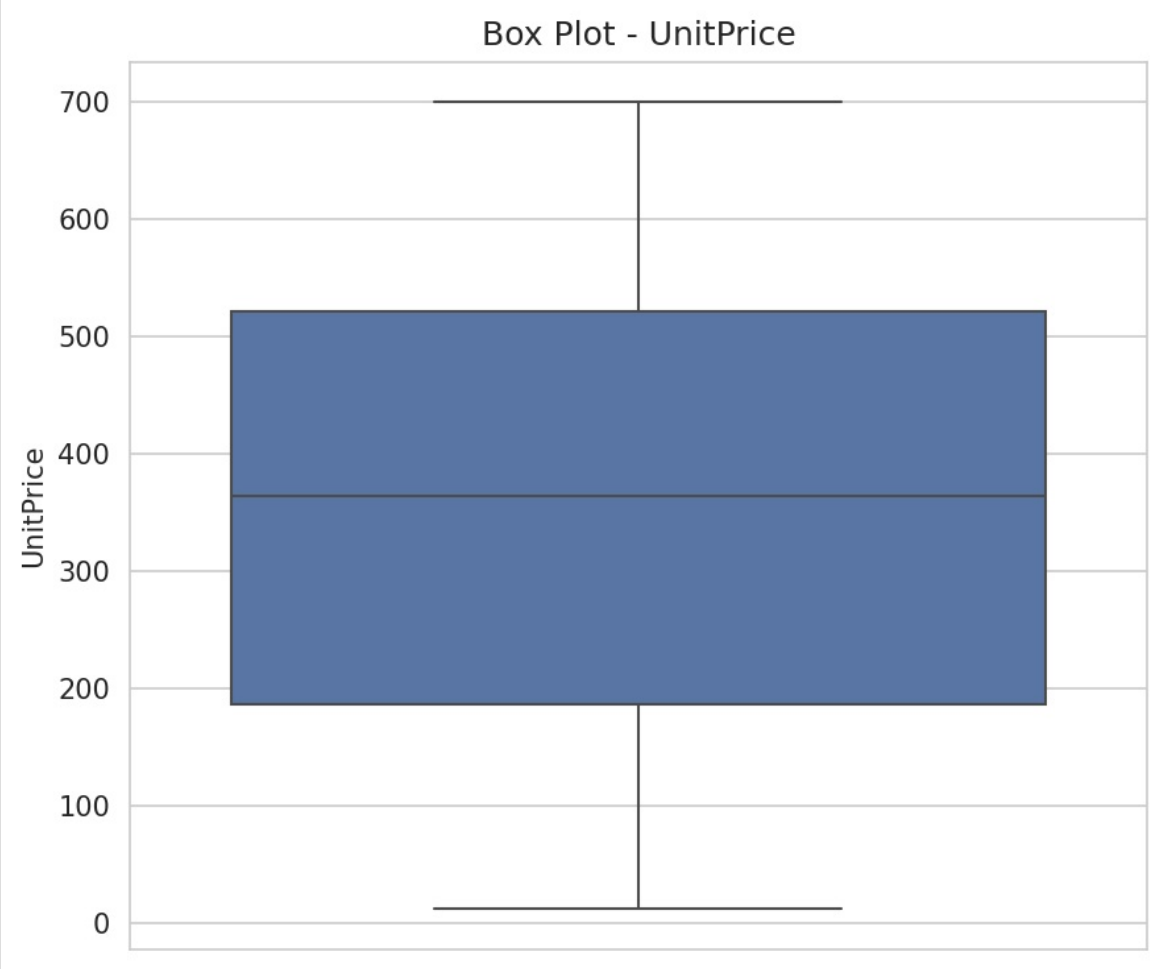
Chart 4 — Box Plot: Quantity

Figure 4 – Distribution Plot (Quantity)

Purpose: To check the distribution and identify outliers in `Quantity`. **Observation:** Quantity values are consistent across orders with no significant outliers. **Business Insight:** Most customers purchase between 1 and 5 units per order, reflecting predictable buying behavior. **Recommendation:** Use this consistent range to plan standard packaging and shipping configurations.

Chart 5 — Box Plot: Unit Price

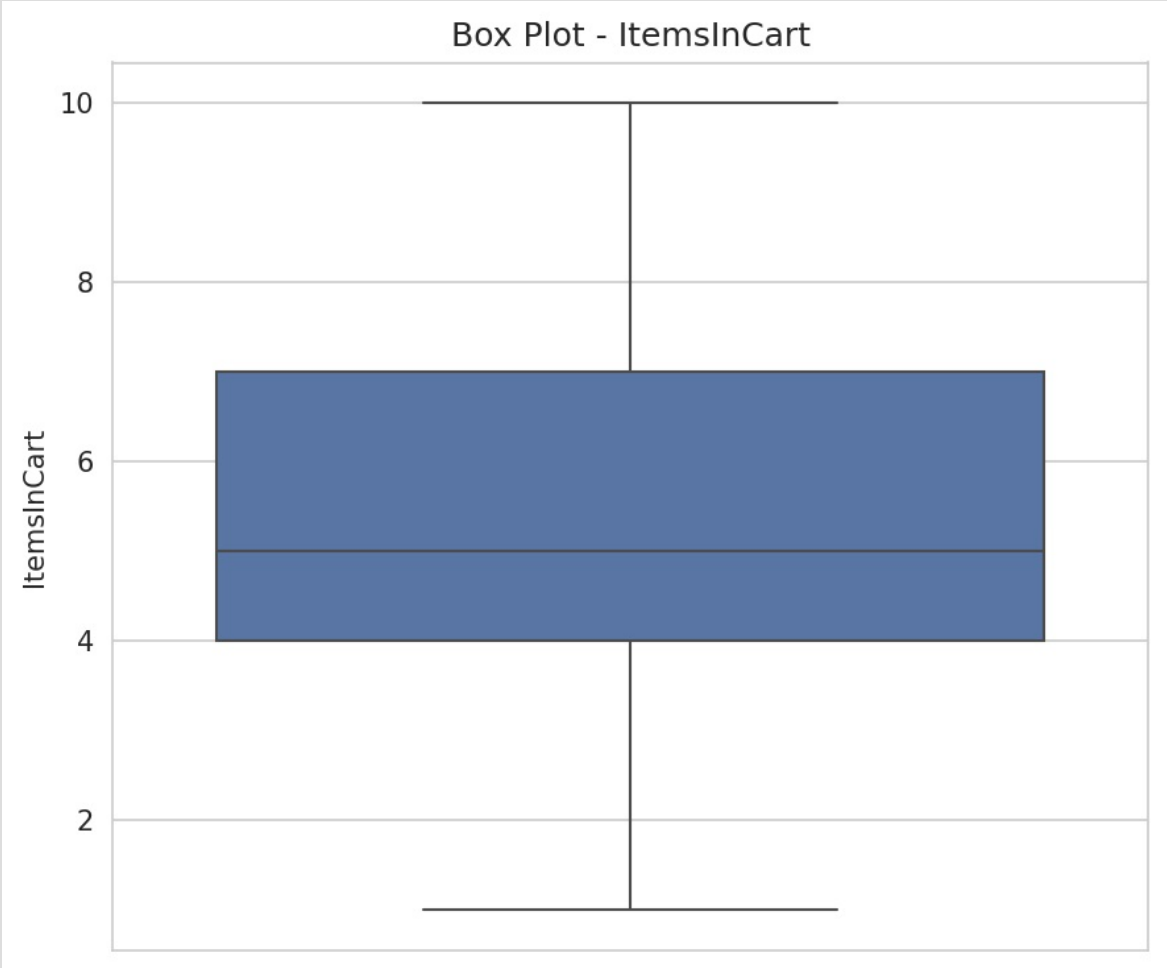
Figure 5 – Distribution Plot (Unit Price)



Purpose: To check the distribution and identify outliers in `UnitPrice`. **Observation:** Product prices are spread across different ranges without extreme outliers. **Business Insight:** The product catalog maintains a balanced pricing structure across categories. **Recommendation:** Maintain the current pricing spread as a baseline for future product or pricing decisions.

Chart 6 — Box Plot: Items In Cart

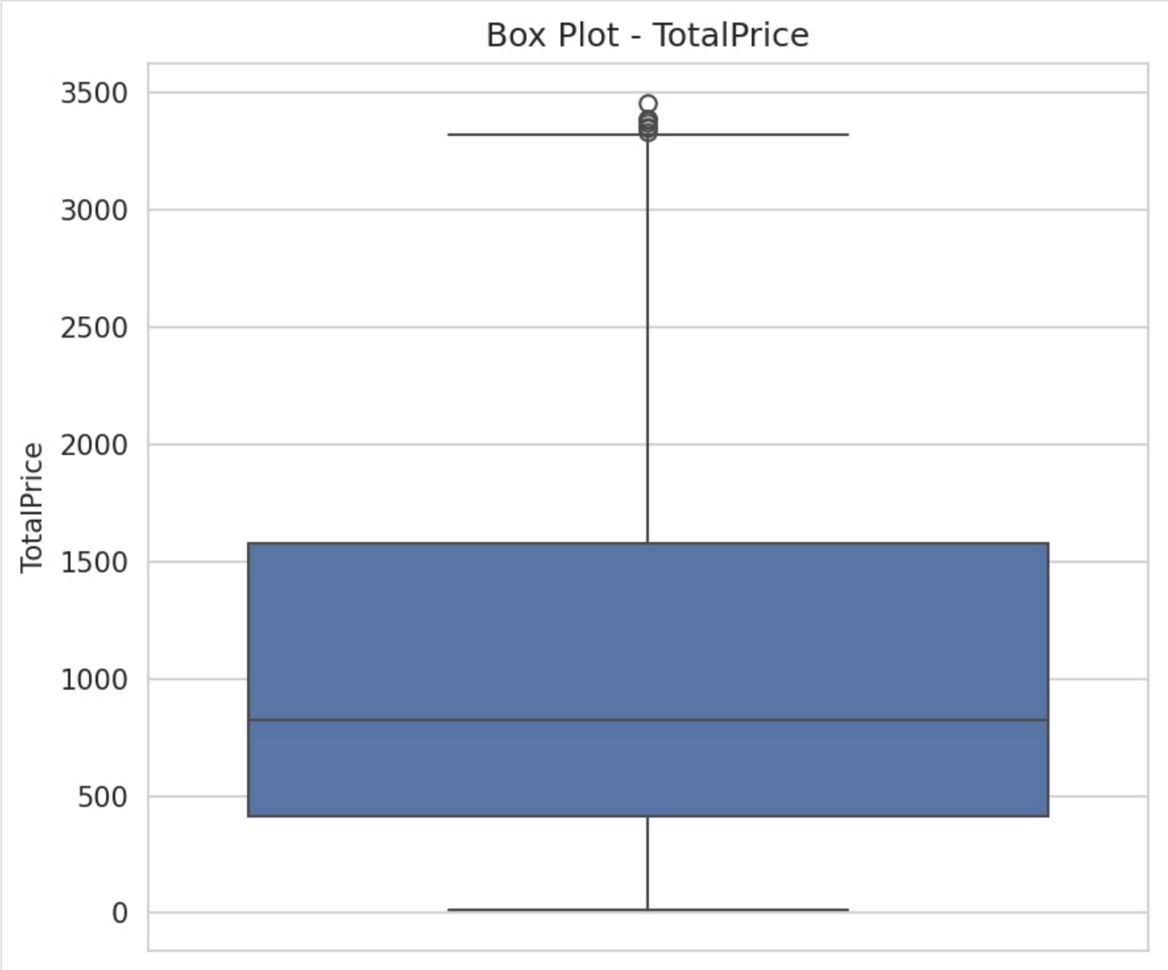
Figure 6 – Distribution Plot (Items in Cart)



Purpose: To check the distribution and identify outliers in `ItemsInCart`. **Observation:** Cart sizes are distributed without extreme outliers. **Business Insight:** Customers generally maintain a consistent cart size when shopping. **Recommendation:** Design cart-based promotions (e.g., incentives for one additional item) around the typical observed cart-size range.

Chart 7 — Box Plot: Total Price

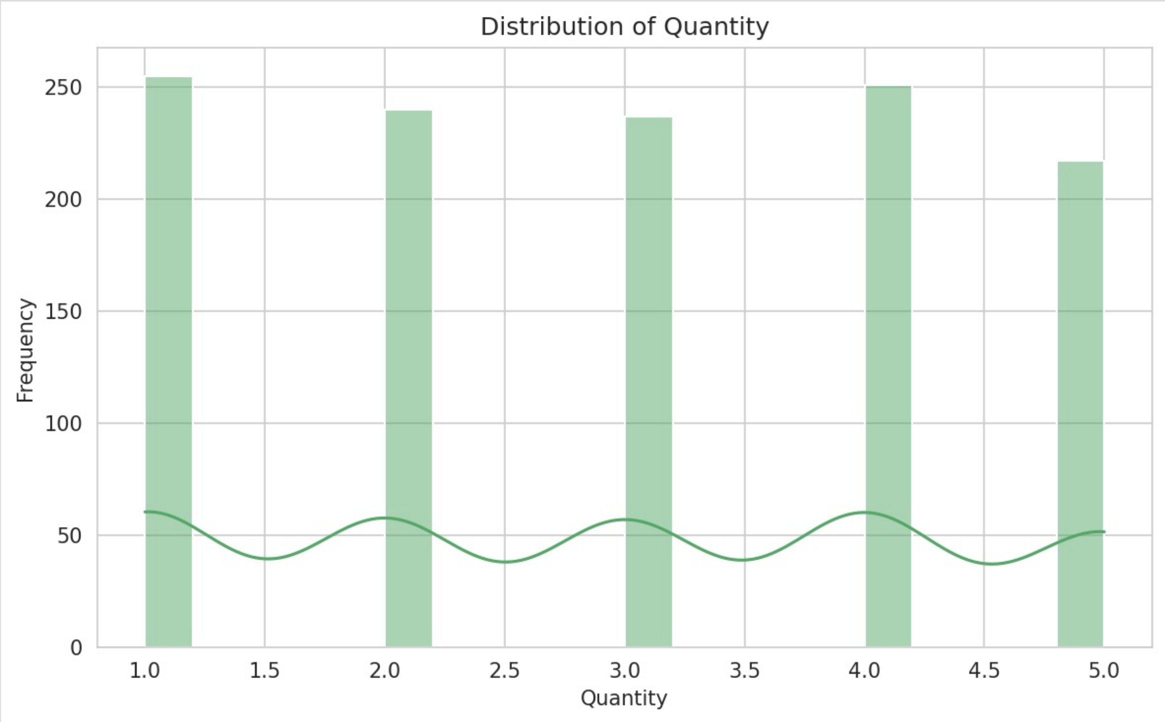
Figure 7 – Distribution Plot (Total Price)



Purpose: To check the distribution and identify outliers in `TotalPrice`. **Observation:** A few high-value orders appear as outliers in the distribution. **Business Insight:** These outlier transactions represent premium or bulk purchases contributing significantly to overall revenue. **Recommendation:** Identify the customers behind these high-value orders and design loyalty or premium-tier programs to retain them.

Chart 8 — Histogram: Quantity

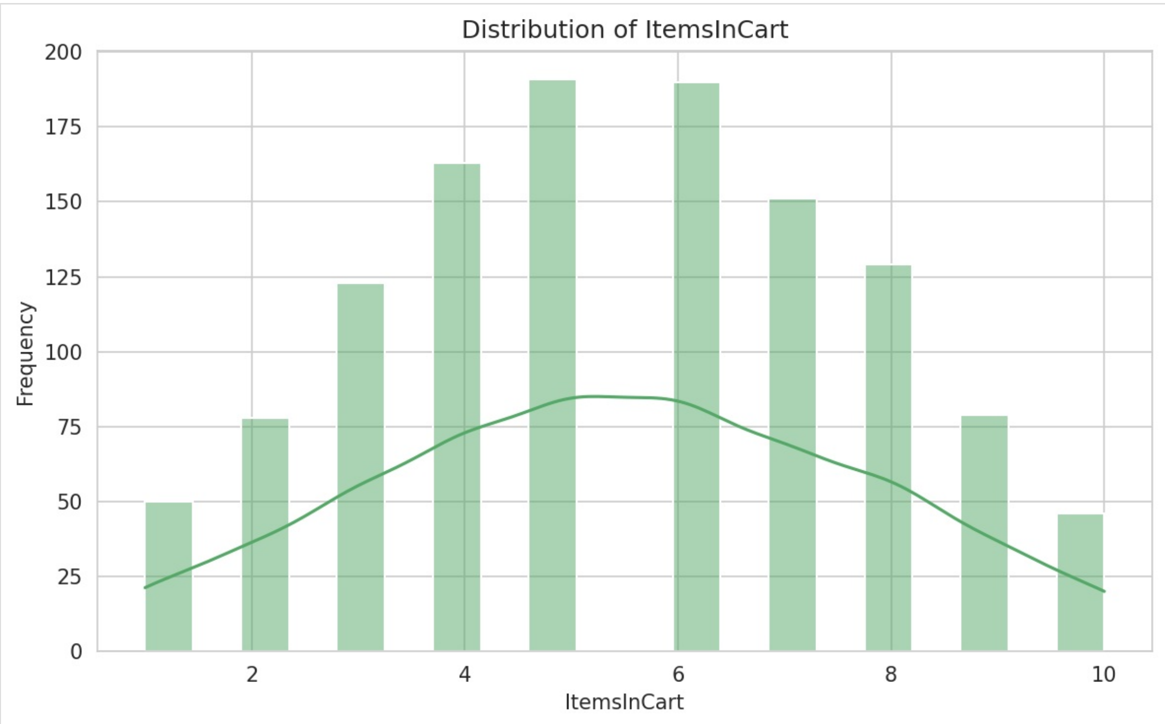
Figure 8 – Frequency Distribution (Quantity)



Purpose: To visualize the frequency distribution of order quantities. **Observation:** Most customers purchase a small to moderate number of items per order. **Business Insight:** There is an opportunity to increase basket size, since most orders fall on the lower end of the quantity range. **Recommendation:** Introduce quantity-based discounts or “buy more, save more” offers.

Chart 9 — Histogram: Items In Cart

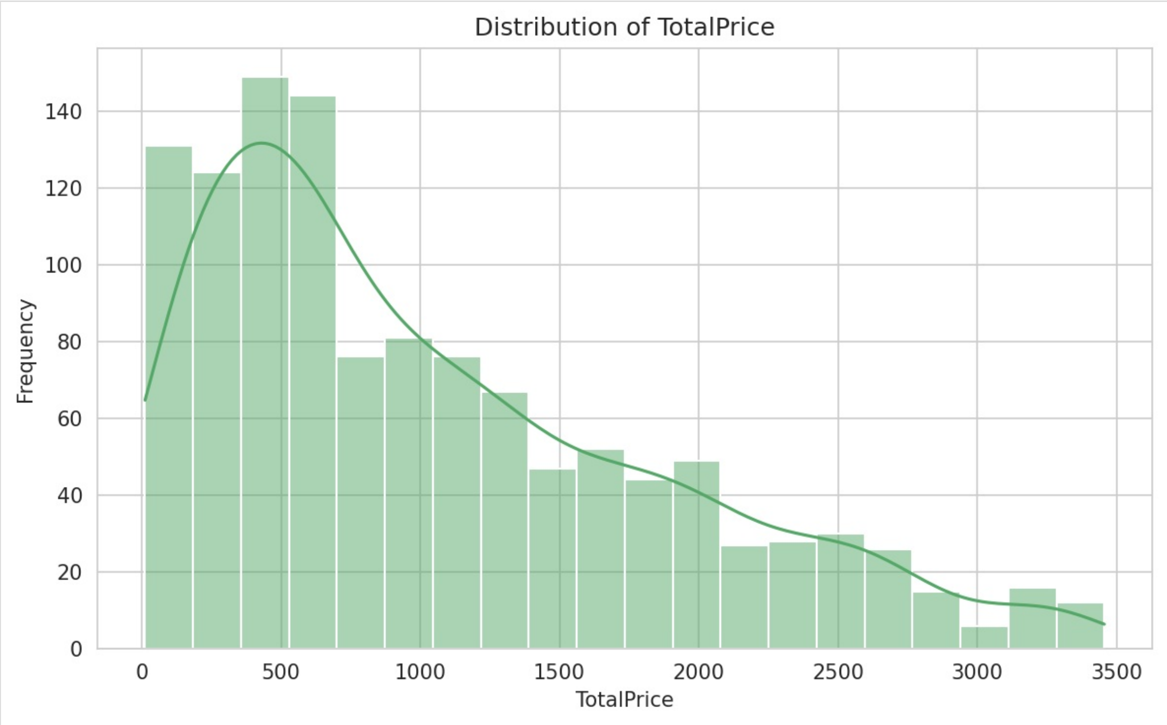
Figure 9 – Frequency Distribution (Items in Cart)



Purpose: To visualize the frequency distribution of items in customers’ carts. **Observation:** Cart sizes are distributed across the 1–10 item range, with most customers maintaining a moderate cart size. **Business Insight:** Cart sizes suggest customers shop with a defined set of needs rather than impulsive bulk additions. **Recommendation:** Use cross-selling suggestions to nudge customers toward slightly larger cart sizes.

Chart 10 — Histogram: Total Price

Figure 10 – Frequency Distribution (Total Price)



Purpose: To visualize the frequency distribution of order values. **Observation:** A limited number of high-value transactions contribute significantly to total revenue. **Business Insight:** Revenue is concentrated among a smaller group of high-spending orders. **Recommendation:** Build a retention strategy specifically focused on high-spending customers.

Chart 11 — Scatter Plot: Quantity vs Total Price

Figure 11 – Quantity vs Total Price

Purpose: To examine the relationship between `Quantity` and `TotalPrice`. **Observation:** A positive relationship is observed; higher quantities generally produce higher order values. **Business Insight:** Quantity meaningfully drives order value, later confirmed with a correlation coefficient of 0.615. **Recommendation:** Promote multi-item purchases through bundling and volume discounts.

Chart 12 — Scatter Plot: Unit Price vs Total Price

Figure 12 – Unit Price vs Total Price

Purpose: To examine the relationship between `UnitPrice` and `TotalPrice`. **Observation:** Higher-priced products tend to contribute to larger order values. **Business Insight:** Product pricing has the strongest influence on order value, later confirmed with a correlation coefficient of 0.717. **Recommendation:** Focus merchandising and promotional efforts on higher-priced product categories.

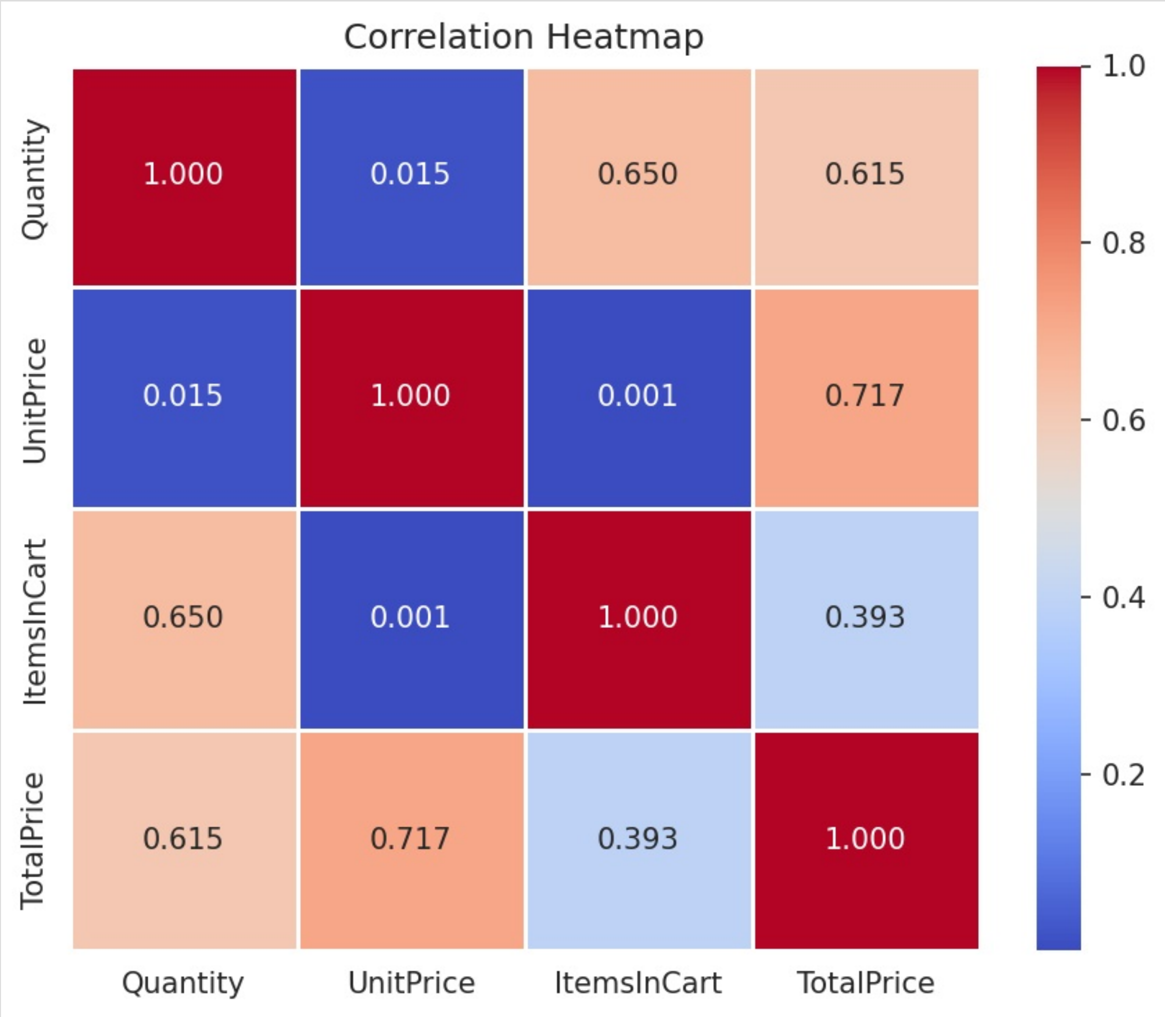
Chart 13 — Scatter Plot: Items In Cart vs Total Price

Figure 13 – Items in Cart vs Total Price

Purpose: To examine the relationship between `ItemsInCart` and `TotalPrice`.
Observation: Customers with more items in their cart generally show higher order values.
Business Insight: Cart size is a useful early indicator of order value, supported by a correlation of 0.393 with `TotalPrice`. **Recommendation:** Monitor cart size in real time and trigger upsell prompts for likely high-value orders.

Chart 14 — Correlation Heatmap

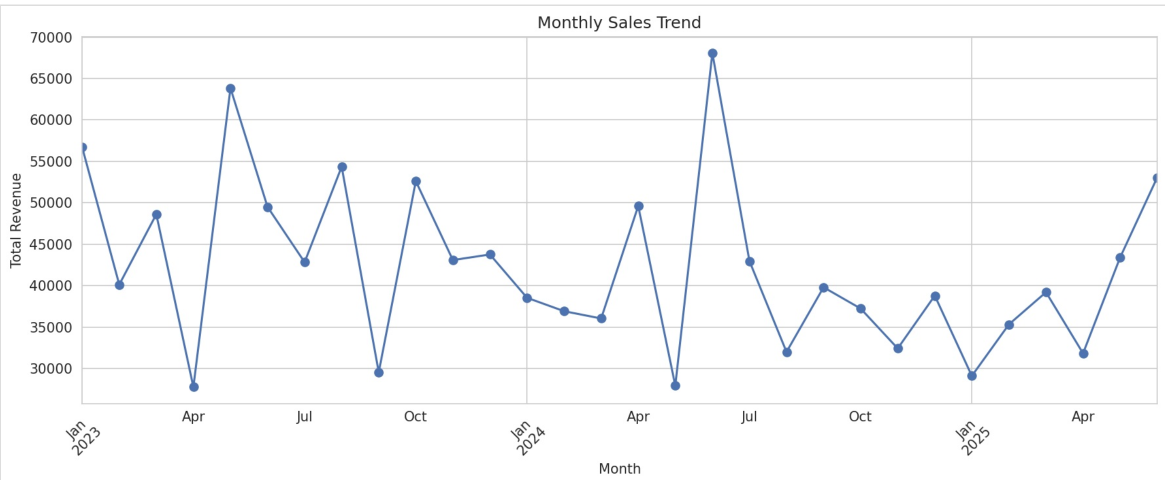
Figure 5 – Correlation Heatmap



Purpose: To visualize the strength and direction of relationships between `Quantity`, `UnitPrice`, `ItemsInCart`, and `TotalPrice`. **Observation:** `UnitPrice` shows the strongest correlation with `TotalPrice` (0.717), followed by `Quantity` (0.615) and `ItemsInCart` (0.393). `ItemsInCart` and `Quantity` show a moderate correlation of 0.650, while `Quantity` and `UnitPrice` are nearly uncorrelated (0.015). **Business Insight:** Revenue is driven primarily by product price, followed by quantity purchased; pricing strategy has greater impact on revenue than any other numeric factor. **Recommendation:** Prioritize pricing strategy and premium product positioning as the primary revenue levers.

Chart 15 — Monthly Sales Trend

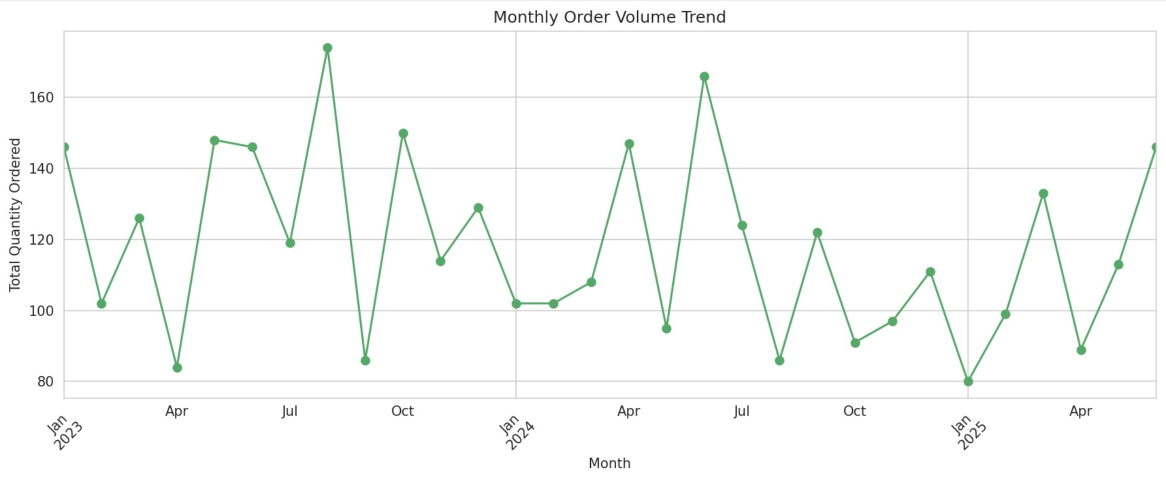
Figure 15 – Monthly Revenue Trend



Purpose: To visualize how total revenue changed on a month-by-month basis. **Observation:** Revenue fluctuates across months, forming an overall trend line from January 2023 to June 2025. **Business Insight:** Tracking monthly revenue helps identify seasonal peaks, slow periods, and overall business trajectory. **Recommendation:** Plan inventory and marketing campaigns around historically stronger months and investigate weaker periods.

Chart 16 — Monthly Order Trend

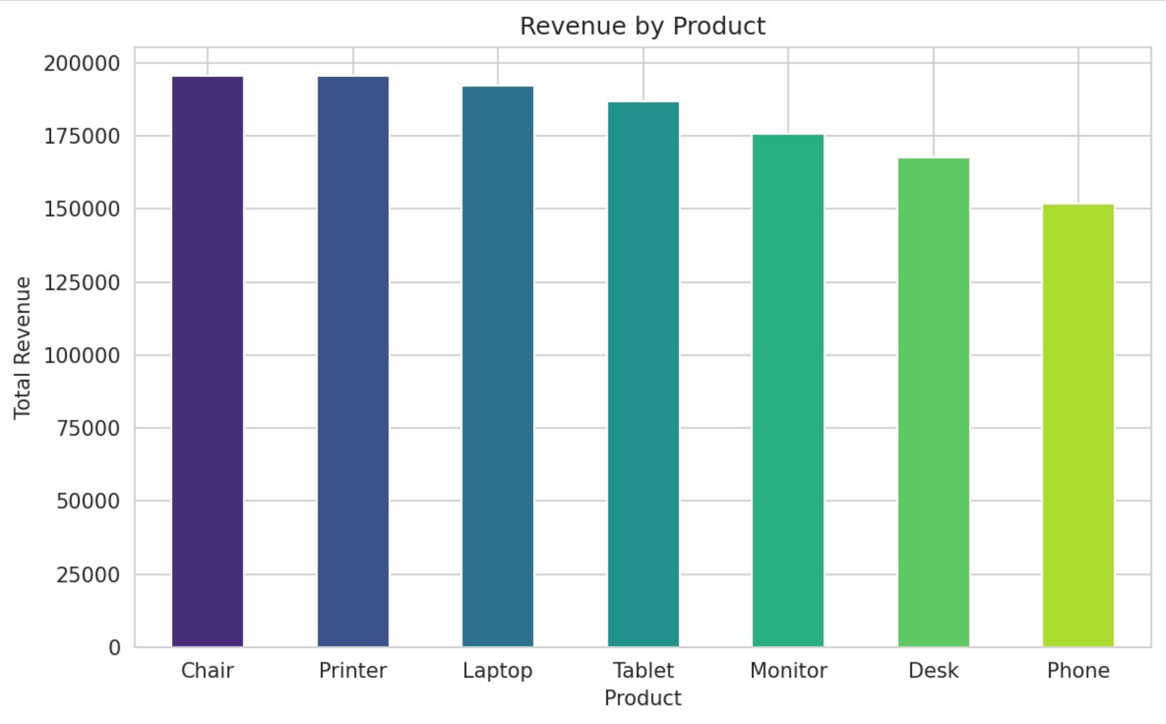
Figure 16 – Monthly Order Volume Trend



Purpose: To visualize how total order quantity changed on a month-by-month basis. **Observation:** The order volume trend follows a pattern similar to the revenue trend across the same period. **Business Insight:** Order volume and revenue moving together suggests revenue changes are primarily driven by volume rather than pricing shifts. **Recommendation:** Align demand planning and inventory levels with the observed monthly order pattern.

Chart 17 — Revenue by Product

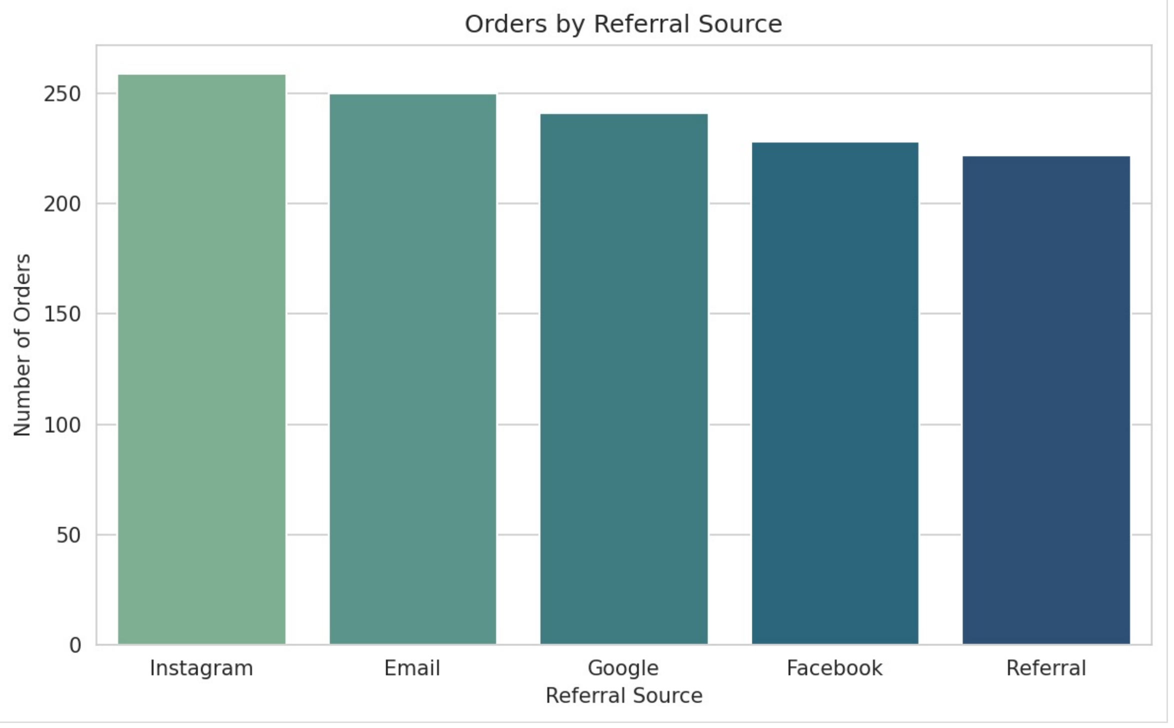
Figure 6 – Revenue by Product Category



Purpose: To compare total revenue generated by each product category. **Observation:** Chair generated the highest revenue (₹195,620), closely followed by Printer and Laptop. Phone generated the lowest revenue (₹151,722). **Business Insight:** Chair leads in revenue and should be prioritized in inventory and promotional planning; Phone underperforms and may need a pricing, marketing, or bundling review. **Recommendation:** Increase stock allocation and marketing focus on Chair, Printer, and Laptop; review Phone’s pricing or bundling strategy.

Chart 18 — Referral Source Analysis

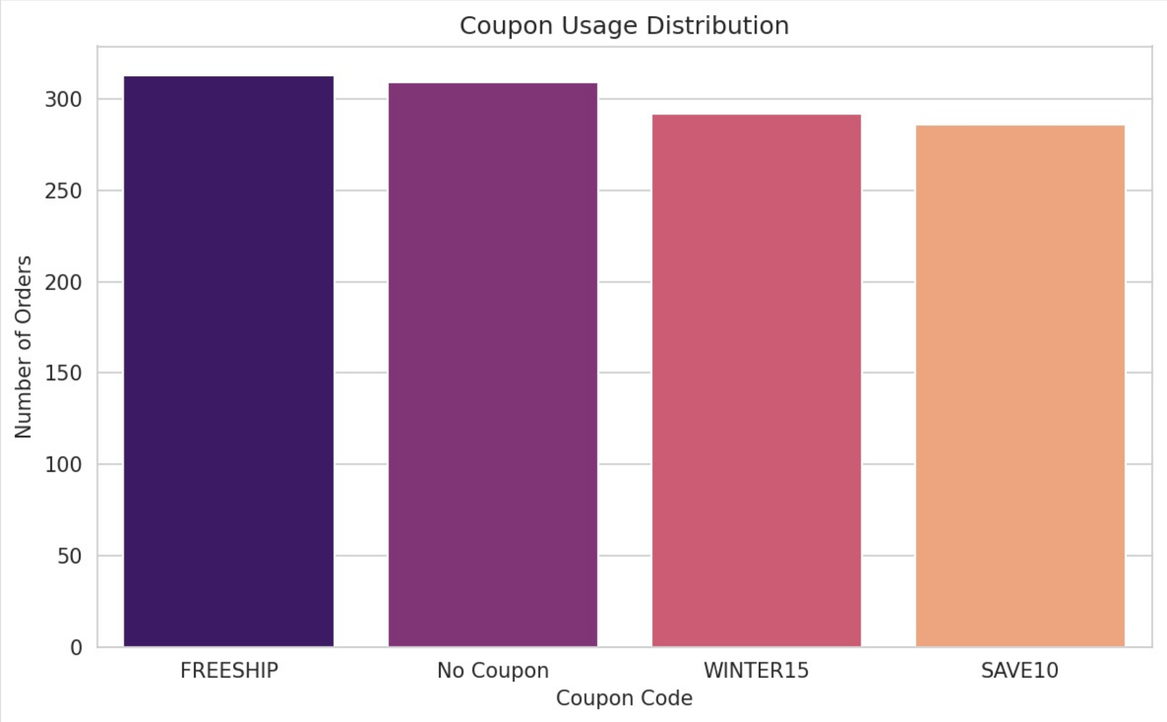
Figure 18 – Customer Acquisition by Referral Source



Purpose: To visualize the distribution of orders by customer acquisition channel. **Observation:** Instagram brought in the most orders (259), while Referral brought in the fewest (222). **Business Insight:** Instagram is the strongest marketing channel and warrants greater budget; the referral program may need stronger incentives. **Recommendation:** Increase investment in Instagram-based marketing and redesign the referral program with stronger incentives.

Chart 19 — Coupon Usage Distribution

Figure 19 – Coupon Usage Distribution



Purpose: To visualize how frequently each coupon code was used across orders. **Observation:** FREESHIP was the most frequently used coupon (313 orders), closely followed by “No Coupon” (309), WINTER15 (292), and SAVE10 (286). **Business Insight:** Free shipping is the most popular incentive among coupon-using customers, while a substantial share of customers complete purchases without any coupon at all. **Recommendation:** Continue offering free shipping as a primary incentive and test expanding its visibility.

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12. Statistical Analysis

Descriptive Statistics

The dataset’s numeric fields were summarized using `df.describe()`:

Metric	Quantity	UnitPrice	ItemsInCart	TotalPrice
Mean	2.95	356.41	5.49	1,053.97
Min	1.00	11.39	1.00	11.39
25%	2.00	186.06	4.00	410.52
50% (Median)	3.00	364.21	5.00	823.62
75%	4.00	521.57	7.00	1,578.48
Max	5.00	699.93	10.00	3,456.40
Std Dev	1.41	197.18	2.28	819.86

Correlation Analysis

Variable Pair	Correlation Coefficient
UnitPrice & TotalPrice	0.717

Quantity & TotalPrice	0.615
ItemsInCart & Quantity	0.650
ItemsInCart & TotalPrice	0.393
Quantity & UnitPrice	0.015

`UnitPrice` has the strongest positive correlation with `TotalPrice`, confirming pricing as the leading driver of revenue, followed by `Quantity`.

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13. SQL Analysis

Not applicable to this project. This notebook-based analysis was performed entirely using Python and Pandas on an in-memory DataFrame; no database, SQL queries, or relational schema were used. SQL analysis on the same underlying dataset is documented separately as Project 3.

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14. Business Insights

1. Chair is the top revenue-generating product (₹195,620), closely followed by Printer and Laptop.
2. Phone is the lowest revenue-generating product (₹151,722), underperforming relative to other categories.
3. Instagram is the most effective acquisition channel, driving 259 orders — the highest among all referral sources.
4. Referral is the weakest acquisition channel, contributing only 222 orders.
5. FREESHIP is the best-performing coupon by revenue, generating ₹335,037 in total sales.
6. Orders without any coupon still perform well, generating ₹322,401 in revenue, indicating healthy organic demand.
7. UnitPrice has the strongest impact on revenue (correlation of 0.717 with TotalPrice).
8. Quantity also significantly drives revenue (correlation of 0.615 with TotalPrice).
9. Cart size influences purchasing behavior (correlation of 0.650 between ItemsInCart and Quantity), suggesting upselling opportunities.
10. Revenue shows a declining trend year-over-year: ₹552,643 (2023) → ₹480,236 (2024) → ₹231,883 (2025, partial year).

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15. Key Findings

- The dataset was clean and well-structured, requiring minimal preprocessing (only `CouponCode` nulls needed handling).
- No duplicate records exist in the dataset.
- Product pricing (`UnitPrice`) is the single strongest driver of order value.
- Chair, Printer, and Laptop form the top revenue tier; Phone is the clear underperformer.
- Instagram outperforms all other referral channels by a meaningful margin.
- Free shipping is the most impactful coupon offer in terms of total revenue.
- Revenue has declined year-over-year from 2023 to 2025 (2025 data is partial, through June).

16. Recommendations

- Prioritize top-performing products (Chair, Printer, Laptop) in inventory planning, homepage placement, and promotional campaigns.
- Re-evaluate Phone’s strategy through price adjustments, bundling with accessories, or a dedicated marketing push.
- Increase marketing investment in Instagram given its strong acquisition performance.
- Strengthen the referral program with better incentives, since it currently underperforms compared to other channels.
- Continue promoting free shipping offers, as FREESHIP is the highest-performing coupon by usage and revenue.
- Leverage pricing strategy as the primary revenue lever, given UnitPrice’s strong correlation with order value.
- Encourage larger order quantities through volume discounts to capitalize on the Quantity–TotalPrice relationship.
- Investigate the declining revenue trend across 2023–2025 to determine whether it reflects seasonality, incomplete 2025 data, or reduced demand.
- Investigate high cancellation and return rates, which together account for a substantial share of total orders.

17. Business Impact

This analysis equips a business with the evidence needed to make resource-allocation decisions rather than relying on intuition. Marketing budget can be directed toward Instagram with greater confidence given its measurable lead in order volume. Inventory and merchandising decisions can prioritize Chair, Printer, and Laptop based on actual revenue contribution rather than assumption. The correlation analysis gives finance and pricing teams a data-backed justification for treating `UnitPrice` as the primary lever for revenue growth. Finally, the year-over-year revenue decline flagged in this analysis gives leadership an early signal to investigate before it compounds further.

18. Challenges Faced

- The `CouponCode` column contained 309 missing values, which required a clear, business-appropriate fill strategy (`"No Coupon"`) before coupon-based analysis could be performed.
- The 2025 revenue figure represents only a partial year (through June 2025), which needed to be considered when interpreting the year-over-year revenue trend rather than treating it as a full annual figure.

19. Limitations

This analysis is descriptive rather than predictive — it identifies patterns and correlations within the existing dataset but does not forecast future revenue or customer behavior. The dataset does not include customer demographic information, product cost/margin data, or marketing spend by channel, which limits the ability to calculate true ROI on acquisition channels or profitability by product. The correlation findings describe association, not causation. The 2025 figures are based on a partial year and should not be compared directly to the full-year totals for 2023 and 2024 without adjustment.

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20. Future Scope

- Build an interactive Power BI or Tableau dashboard on top of this analysis for real-time monitoring.
- Extend the year-over-year trend analysis once full 2025 data becomes available.
- Apply customer segmentation (e.g., RFM analysis) to identify high-value customer groups.
- Explore time-series forecasting to predict future revenue trends.
- Investigate the drivers behind high cancellation and return rates using additional operational data not present in the current dataset.

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21. Conclusion

This project demonstrates how exploratory data analysis can convert raw transactional data into clear, decision-ready insights using Python’s core data science stack. By combining distribution analysis, outlier detection, correlation analysis, and trend analysis, the project identifies specific, evidence-based opportunities for product strategy, marketing budget allocation, and revenue growth, providing a solid foundation for further analytics work such as dashboarding and forecasting.

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22. Project Folder Structure

```
Project-2-Python-EDA/
│
├── Dataset/
│   └── Dataset_for_Data_Analytics.xlsx
│
├── Notebook/
│   └── DecodeLab_project_2.ipynb
│
├── Images/
│   ├── chart01.png
│   ├── chart02.png
│   ├── ...
│   └── chart19.png
│
├── README.md
├── requirements.txt
└── Documentation/
    └── Project_Documentation.pdf
```

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23. GitHub Repository Structure

```
repository-root/
│
├── README.md
├── requirements.txt
├── data/
│   └── Dataset_for_Data_Analytics.xlsx
├── notebooks/
│   └── DecodeLab_project_2.ipynb
├── images/
│   └── (exported chart screenshots)
└── docs/
    └── Project_Documentation.pdf
```

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24. Installation Guide

1. Install Python 3.x from the official Python website if not already installed.
2. Install Jupyter Notebook: `pip install notebook`
3. Install the required analysis libraries:

   ```
   pip install pandas numpy matplotlib seaborn
   ```

4. Clone or download the project repository to your local machine.

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25. How to Run the Project

1. Navigate to the project folder in a terminal.
2. Launch Jupyter Notebook: `jupyter notebook`
3. Open `DecodeLab_project_2.ipynb`.
4. Run all cells sequentially from top to bottom to reproduce the data cleaning, visualizations, and statistical analysis.
5. Exported chart images can be saved to the `images/` folder for use in documentation.

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26. File Description

File	Description
`Dataset_for_Data_Analytics.xlsx`	The raw e-commerce order dataset (1,200 records, 14 columns) used as the input for this analysis
`DecodeLab_project_2.ipynb`	The Jupyter Notebook containing all Python code, cleaning steps, visualizations, and statistical analysis

`README.md`	Project README summarizing the analysis for GitHub viewers
`requirements.txt`	List of Python libraries required to run the notebook

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27. Screenshots Section

Figure	Caption
Figure 1	Dataset Overview (head/info output)
Figure 2	Data Cleaning (missing value and duplicate check output)
Figure 3	Payment Method Distribution Plot
Figure 4	Distribution Plot (Box Plots — Quantity, UnitPrice, ItemsInCart, TotalPrice)
Figure 5	Correlation Heatmap
Figure 6	Business Insights (Revenue by Product chart)
Figure 7	Final Results (Monthly Sales Trend)

Note: Actual screenshots from the notebook should be inserted in place of these placeholders before final submission.

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28. Skills Demonstrated

Technical Skills: Python programming, Pandas data manipulation, NumPy numerical operations, data visualization with Matplotlib and Seaborn, descriptive statistics, correlation analysis, time-series aggregation, Jupyter Notebook workflow.

Professional Skills: Structuring an end-to-end analysis from raw data to business recommendation, translating statistical findings into business language, identifying and prioritizing actionable insights, documenting analytical work for non-technical stakeholders.

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29. Learning Outcomes

Through this project, practical experience was gained in applying the full EDA workflow to a real transactional dataset — from initial data quality checks through to multi-chart visual analysis and correlation-based insight generation. The project reinforced how to translate a statistical result (such as a correlation coefficient) into a specific, defensible business recommendation, and how to structure a notebook-based analysis so that every chart serves a clear analytical purpose rather than being produced for its own sake.

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30. Industry Relevance

This project reflects a task structure commonly assigned to junior and entry-level Data Analysts: given a transactional dataset, identify quality issues, profile the data, and surface insights that inform marketing, inventory, and pricing decisions. The combination of distribution analysis, correlation analysis, and trend analysis mirrors the standard first-pass analytical workflow used in real e-commerce and retail analytics teams before any dashboard or model is built.

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31. Recruiter Highlights

This project demonstrates the ability to independently take a raw dataset through a complete, structured EDA pipeline using Python's core data science libraries, and to consistently translate every chart and statistic into a specific business recommendation rather than a generic observation. It shows comfort with Pandas-based data cleaning, multiple visualization types (distribution, relationship, and trend charts), and correlation-based reasoning — all of which are directly transferable to a Data Analyst role.

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32. Appendix

Python Code References

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Data Dictionary

See Section 6 ("Dataset Description") for the complete column-by-column data dictionary.

Glossary

- **EDA (Exploratory Data Analysis):** The process of visually and statistically examining a dataset to summarize its main characteristics before formal modeling.
- **Correlation Coefficient:** A statistical measure (ranging from -1 to 1) describing the strength and direction of a linear relationship between two numeric variables.
- **Outlier:** A data point that differs significantly from other observations in a dataset.

References

- Dataset: `Dataset_for_Data_Analytics.xlsx` (project-provided file)
- Notebook: `DecodeLab_project_2.ipynb` (project-provided file)